

Chapter 3

Z Transform

CHAPTER HIGHLIGHTS

- 🔊 Sampling
- 🔊 Sampling Theorem
- 🔊 The z-transform
- 🔊 Region of Convergence
- 🔊 z-transform Properties
- 🔊 Inverse z-transform
- 🔊 Analysis of Discrete-time LTI Systems
- 🔊 Characterization of Discrete Time LTI Systems
- 🔊 The Unilateral z-transform
- 🔊 Relationship between z and Laplace Transform
- 🔊 Structures for Realization of IIR Systems

SAMPLING

Let $x(t)$ be any continuous signal periodically sampled at equal sample time period T_s to get discrete-time signal represented by $x(n)$



$$x(n) = x(nT_s)$$

$$n = 0, \pm 1, \pm 2, \pm 3 \dots \infty$$

T_s – sampling time period

Sampling Theorem

Let $x(t)$ is strictly band-limited signal with Fourier spectrum. $X(j\omega) = 0$ for $|\omega| > \omega_m$

Then, $x(t)$ can be recovered without loss of information from its sample signal $x(n) = x(nT_s)$, if the sampling frequency ω_s is greater than twice of the maximum frequency ω_m (of $x(t)$)

$$\omega_s > 2\omega_m$$

Minimum sampling frequency $\omega_s = 2\omega_m$ is called Nyquist sampling rate.

In case of sampling frequency $\omega_s < 2\omega_m$ $X_p(j\omega)$ which is spectrum of $x_p(t) = x(nT_s)$ no longer contains all information of $X(j\omega)$; hence, $x(t)$ is not fully recoverable from sampler $x_p(t)$, and there is definite loss of original signal $x(t)$.

The loss of information in such sampling case ($\omega_s < 2\omega_m$) is due to super imposition of high-frequency components onto the less-frequency component. This phenomenon is called frequency folding or aliasing.

Due to aliasing, high-frequency noise components get mixed with signal frequencies between 0 to $\omega_s/2$.

To prevent high-frequency noise components having frequency higher than $\omega_s/2$ an analogue filter is used before sampler to attenuate signals with frequency higher than $\omega_s/2$. Such filters are called antialiasing filters.

Z TRANSFORM

A discrete-time LTI system with impulse response $h(n)$, the output $y(n)$ of the system to the complex exponential input of the form z^n is $y(n) = T\{z^n\}$

$$\begin{aligned}
 &= h(n) \times z^n \\
 &= \sum_{k=-\infty}^{\infty} h(k) z^{n-k} \\
 &= z^n \sum_{k=-\infty}^{\infty} h(k) z^{-k} , \\
 &= H(z) z^n
 \end{aligned}$$

$$\text{where } H(z) = \sum_{n=-\infty}^{\infty} h(n) z^{-n}$$

The variable z is generally complex valued.

When $|z|$ is not restricted to unity, the summation is referred to as the z transform of $h(n)$.

Definition

For general discrete-time signal $x(n)$, the z transfer function $X(z)$ is defined as follows:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

The variable is generally complex valued and is expressed in polar form as $z = re^{j\Omega}$.

Where ' r ' is the magnitude of z and ' Ω ' is the angle of ' z '. $x(n)$ and $X(z)$ are said to form a z transform pair as $x(n) \leftrightarrow X(z)$

REGION OF CONVERGENCE

The range of values of the complex variable z for which the z transform converges is called the region of convergence. Consider the sequence $x(n) = a^n u(n)$

$$\begin{aligned} \text{Z transform of } x(n), X(z) &= \sum_{n=-\infty}^{\infty} a^n u(n) z^{-n} \\ &= \sum_{n=0}^{\infty} (az^{-1})^n \end{aligned}$$

For the convergence, this summation should be finite.

This ROC is the range of values of z for which $|az^{-1}| < 1$, or $|z| > |a|$ then

$$X(z) = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}}, |z| > |a| = \frac{z}{z - a}, |z| > |a|.$$

There is one zero, at $z = 0$, and one pole at $z = a$, ROC and the pole zero plot for the examples are shown in figure.

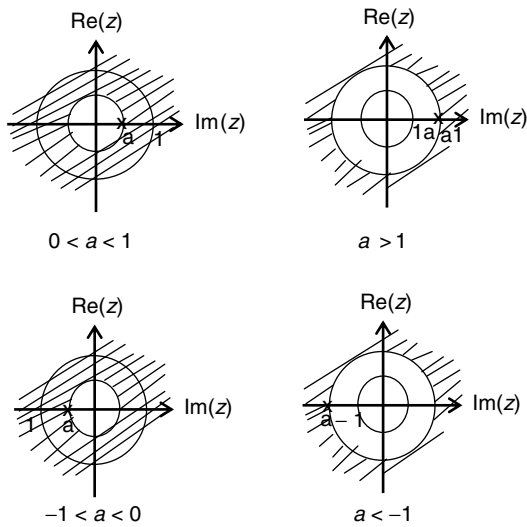


Figure 1 ROC of the form $|z| > |a|$.

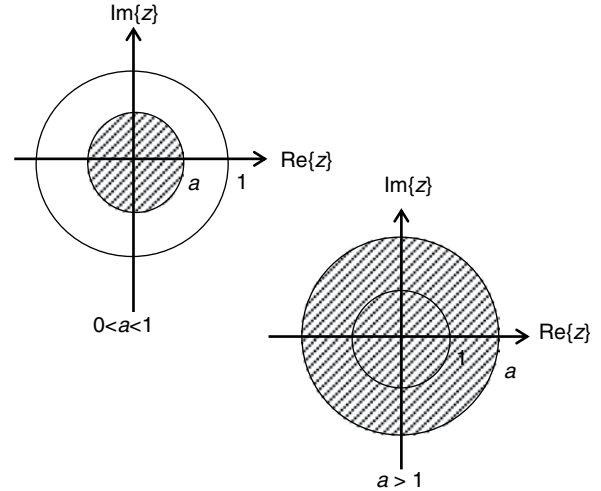
Now let $x(n) = -a^n u(-n - 1)$, then z transform $X(z)$

$$\begin{aligned} &= - \sum_{n=-\infty}^{+\infty} a^n u(-n - 1) z^{-n} = - \sum_{n=-\infty}^{-1} (az^{-1})^n \\ &= - \sum_{n=1}^{\infty} (a^{-1}z)^n. \text{ By taking } n = -n, \\ &= 1 - \sum_{n=0}^{\infty} (a^{-1}z)^n \end{aligned}$$

If $|a^{-1}z| < 1$, then the above summation converges, i.e., for $|z| < |a|$

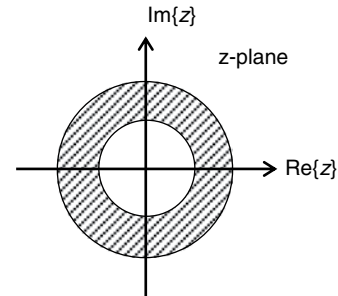
$$X(z) = 1 - \frac{1}{1 - a^{-1}z} = \frac{-a^{-1}z}{1 - a^{-1}z} = \frac{z}{z - a}, |z| < |a|$$

The pole zero plots are

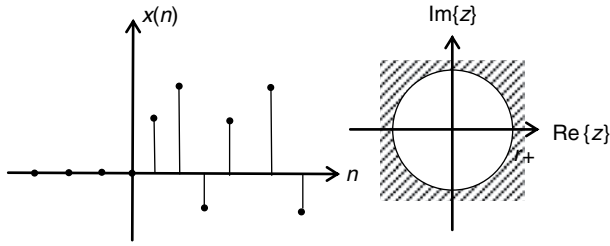


Properties of ROC

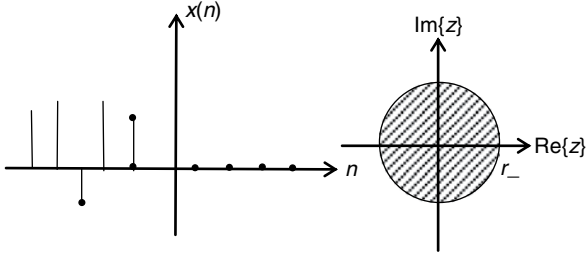
1. The ROC of $X(z)$ consists of a ring in the z plane centred about the origin. In some cases, the inner boundary can extend inward to the origin, in which case the ROC becomes a disc. In other case, the outer boundary can extend outward to infinity.



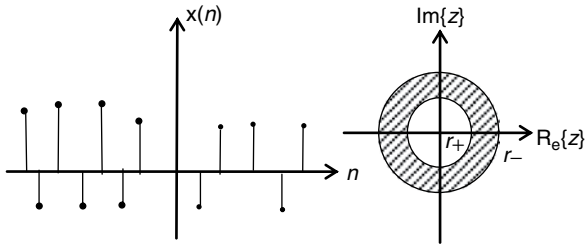
2. ROC does not contain any poles; At poles, $X(z)$ is infinite, and therefore, by definition, it does not converge
3. If $x(n)$ is a finite sequence, then $X(z)$ converges for every value of z , then ROC is the entire z plane except possibly $z = 0$ and/ or $z = \infty$
4. If $x(n)$ is a right-handed sequence, ($x(n) = 0$, for $n < N_1 < \infty$), then ROC is of the form $|z| > r_{\max}$ or $\infty > |z| > r_{\max}$ where $r_{\max}$ is the largest magnitude of any of the poles of $X(z)$. Thus, ROC is the exterior of the circle $|z| = r_{\max}$
5. If $x(n)$ is a left-handed sequence ($x(n) = 0$, for $n > N_2 > -\infty$), then ROC is of the form $|z| < r_{\min}$ or $0 < |z| < r_{\min}$ where $r_{\min}$ is the smallest magnitude of any of the poles of $X(z)$, and thus, ROC is interior of the circle $|z| = r_{\min}$
6. If $x[n]$ is two-sided sequence, ($x(n)$ is infinite duration sequence). Then, the ROC is of the form $r_1 < |z| < r_2$ where r_1, r_2 are the magnitudes of the two poles of $X(z)$. It forms like a ring.
7. Relationship between the ROC and the time extent of a signal



A right-sided signal has an ROC of form $|z| > r_+$



A left-sided signal has an ROC of form $|z| < r_-$



A two-sided signal has an ROC of form $r_+ < |z| < r_-$

Z TRANSFORMS OF THE SOME COMMON SEQUENCES

For unit impulse signal $\delta[n]$, its z transform is given by $\delta(n)$

$$\xrightarrow{z} \sum_{n=-\infty}^{\infty} \delta(n) z^{-n} = 1$$

ROC is entire z plane, including $z = 0$, and $z = \infty$

For delayed unit impulse

$$\delta(n - n_0) \xrightarrow{z} \sum_{n=-\infty}^{\infty} \delta(n - n_0) z^{-n} = z^{-n_0}$$

ROC is entire z plane, except $z = 0$

For advanced unit impulse

$$\delta(n + n_0) \xrightarrow{z} \sum_{n=-\infty}^{\infty} \delta(n + n_0) z^{-n} = z^{n_0}$$

ROC is entire z plane, except $z = \infty$

$x(n)$	$X(z)$	ROC
$\delta(n)$	1	All z
$u(n)$	$\frac{1}{1 - z^{-1}}$	$ z > 1$

(Continued)

$-u(-n - 1)$	$\frac{1}{1 - z^{-1}}$	$ z < 1$
$\delta(n - m)$	z^{-m}	All z except 0 ($m > 0$) or ∞ ($m < 0$)
$a^n u(n)$	$\frac{1}{1 - az^{-1}}$	$ z > a $
$-a^n u(-n - 1)$	$\frac{1}{1 - az^{-1}}$	$ z < a $
$na^n u(n)$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $
$-na^n u(-n - 1)$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z < a $
$(n + 1)a^n u(n)$	$\frac{1}{(1 - az^{-1})^2}$	$ z > a $
$(\cos \Omega_0 n) u(n)$	$\frac{z^2 - (\cos \Omega_0)z}{z^2 - (2 \cos \Omega_0)z + 1}$	$ z > 1$
$(\sin \Omega_0 n) u(n)$	$\frac{(\sin \Omega_0)z}{z^2 - (2 \cos \Omega_0)z + 1}$	$ z > 1$
$(r^n \cos \Omega_0 n) u(n)$	$\frac{z^2 - (r \cos \Omega_0)z}{z^2 - (2r \cos \Omega_0)z + r^2}$	$ z > r$
$(r^n \sin \Omega_0 n) u(n)$	$\frac{(r \sin \Omega_0)z}{z^2 - (2r \cos \Omega_0)z + r^2}$	$ z > r$
$a^n \quad 0 < n < N - 1$	$\frac{1 - a^N z^{-N}}{1 - az^{-1}}$	$ z > 0$
$0 \quad \text{otherwise}$		

Solved Examples

Example 1

What are the ROC and z transforms of the signal $x(n) = 5$

$$\left(\frac{1}{4}\right)^n u(n) - 4 \left(\frac{1}{3}\right)^n u(n)?$$

Solution

$$\begin{aligned}
 X(z) &= \sum_{n=-\infty}^{\infty} \left\{ 5 \left(\frac{1}{4}\right)^n u(n) - 4 \left(\frac{1}{3}\right)^n u(n) \right\} z^{-n} \\
 &= 5 \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n z^{-n} - 4 \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n z^{-n} \\
 &= 5 \sum_{n=0}^{\infty} \left(\frac{1}{4} z^{-1}\right)^n - 4 \sum_{n=0}^{\infty} \left(\frac{1}{3} z^{-1}\right)^n \\
 &= \frac{5}{1 - \frac{1}{4} z^{-1}} - \frac{4}{1 - \frac{1}{3} z^{-1}} = \frac{1 - \frac{5}{3} z^{-1} + z^{-1}}{\left(1 - \frac{1}{4} z^{-1}\right) \left(1 - \frac{1}{3} z^{-1}\right)} \quad (1)
 \end{aligned}$$

$$= \frac{1 - \frac{2}{3}z^{-1}}{\left(1 - \frac{1}{4}z^{-1}\right)\left(1 - \frac{1}{2}z^{-1}\right)} = \frac{z\left(z - \frac{2}{3}\right)}{\left(z - \frac{1}{4}\right)\left(z - \frac{1}{2}\right)}$$

For the convergence of $X(z)$, both sums in the equation (1) should converge, which requires $\left|\frac{1}{4}z^{-1}\right| < 1$, and $\left|\frac{1}{3}z^{-1}\right| < 1$ or equivalently $|z| > \frac{1}{4}$ and $|z| > \frac{1}{3}$

By combining these two regions, the ROC will be $|z| > \frac{1}{3}$

Example 2

What is the z transform and ROC of the signal

$$x(n) = \begin{cases} a^n & 0 \leq n \leq N-1, a > 0 \\ 0 & \text{otherwise} \end{cases}$$

Solution

$$\begin{aligned} X(z) &= \sum_{n=0}^{N-1} a^n z^{-n} = \sum_{n=0}^{N-1} (az^{-1})^n \\ &= \frac{1 - (az^{-1})^N}{1 - (az^{-1})} = \frac{1}{z^{N-1}} \cdot \frac{z^N - a^N}{z - a} \\ &= \frac{1}{z^{N-1}} \frac{(z-a)(a^{N-1} + K_1 z^{N-2} + \dots + a^{N-1})}{z - a} \\ &= \frac{1}{z^{N-1}} (z^{N-1} + K_1 z^{N-2} + \dots + a^{N-1}) \end{aligned}$$

There is a pole of order $N-1$ at $z=0$, and ROC is all values of z except $z=0$

Example 3

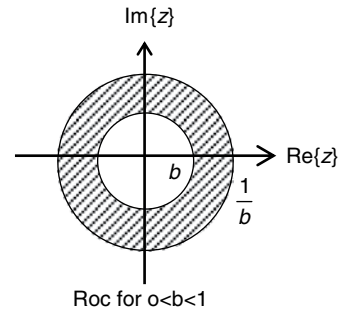
What is the z transform of $x(n) = b^{|n|}$ $b > 0$

Solution

$$\begin{aligned} x(n) &= b^n u(n) + b^{-n} u(-n-1) \\ b^n u(n) &\xrightarrow{z} \frac{1}{1 - bz^{-1}}, |z| > b, \\ \left(\frac{1}{b}\right)^n u(-n-1) &\xrightarrow{z} \frac{-1}{1 - b^{-1}z^{-1}}, |z| < \frac{1}{b} \end{aligned}$$

When $b > 1$, there is no common ROC for the above functions, But for $0 < b < 1$, the ROCs of above two equations will overlap, and thus, the z transform for composite sequence is

$$\begin{aligned} X(z) &= \frac{1}{1 - bz^{-1}} - \frac{1}{1 - b^{-1}z^{-1}}, b < |z| < \frac{1}{b} \\ &= \frac{z}{z-b} - \frac{z}{z-b^{-1}}, b < |z| < \frac{1}{b} \\ &= \frac{b^2 - 1}{b} \frac{z}{(z-b)(z-b^{-1})}, b < |z| < \frac{1}{b} \end{aligned}$$



Z TRANSFORM PROPERTIES

1. Linearity

$$\begin{aligned} x_1(n) &\leftrightarrow X_1(z), \text{ ROC} = R_1 \\ x_2(n) &\leftrightarrow X_2(z), \text{ ROC} = R_2 \\ a_1 x_1(n) + a_2 x_2(n) &\leftrightarrow a_1 X_1(z) + a_2 X_2(z) \\ R^1 &\supset R_1 \cap R_2 \end{aligned}$$

2. Time shifting

$$\begin{aligned} x_1(n) &\leftrightarrow X(z), \text{ ROC} = R \\ x(n - n_0) &\leftrightarrow z^{-n_0} X(z), R^1 = R \cap \{0 < |z| < \infty\} \end{aligned}$$

3. Multiplication by z_0^n

$$\begin{aligned} z_0^n x(n) &\leftrightarrow X\left(\frac{z}{z_0}\right), R^1 = |z_0| R \\ a^n x(n) &\leftrightarrow X(a^{-1} z), R^1 = |a| R \\ e^{+j\Omega_0 n} x(n) &\leftrightarrow X(e^{-j\Omega_0 n}), R^1 = R \end{aligned}$$

4. Time reversal

$$x(-n) \leftrightarrow X\left(\frac{1}{z}\right), R^1 = \frac{1}{R}$$

5. Multiplication by 'n'

$$nx(n) \leftrightarrow -z \frac{dX(z)}{dz}, R^1 = R$$

6. Accumulation

$$\sum_{k=-\infty}^n x(k) \leftrightarrow \frac{1}{1 - z^{-1}} X(z), R^1 \supset R \cap \{|z| > 1\}$$

7. Convolution

$$x_1(n) \times x_2(n) \leftrightarrow X_1(z) X_2(z), R^1 \supset R_1 \cap R_2$$

8. Conjugation

$$x^*(n) \leftrightarrow X^*(z^*) \text{ with ROC} = R$$

9. First difference

$$\begin{aligned} x(n) - x(n-1) &\leftrightarrow (1 - z^{-1})X(z), \\ R^1 \cap R &\supset \{|z| > 0\} \end{aligned}$$

10. Initial value theorem

$$\text{if } x(n) = 0, \text{ for } n < 0, \text{ then } x(0) = \lim_{z \rightarrow \infty} z X(z)$$

11. Final value theorem

$$\lim_{n \rightarrow \infty} x(n) = \lim_{z \rightarrow 1} (z-1) X(z), \text{ Poles of } (z-1) X(z) \text{ must lie inside the unit circle.}$$

12. Time expansion

$$x_k(n) = x(n/k) \leftrightarrow X(z^k), R^1 = R^{1/k}, \text{ for } n, k, \text{ and } \frac{n}{k} \text{ are integers}$$

INVERSE Z TRANSFORM

Inversion of the z transform to find the sequence $x(n)$ from its z transform $X(z)$ is called inverse z transform.

$$x(n) = z^{-1} \{X(z)\}. \quad x(n) = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz$$

Where C is a counter clockwise contour of integration enclosing the origin

Example 4

Consider the z transform $X(z)$

$$X(z) = \frac{3 - \frac{11}{15}z^{-1}}{\left(1 - \frac{1}{3}z^{-1}\right)\left(1 - \frac{1}{5}z^{-1}\right)}$$

Find the inverse z transform for different ROCs

(i) $|z| > \frac{1}{3}$ (ii) $\frac{1}{5} < |z| < \frac{1}{3}$ (iii) $|z| < \frac{1}{5}$

Solution

$$X(z) = \frac{3 - \frac{11}{15}z^{-1}}{\left(1 - \frac{1}{3}z^{-1}\right)\left(1 - \frac{1}{5}z^{-1}\right)}$$

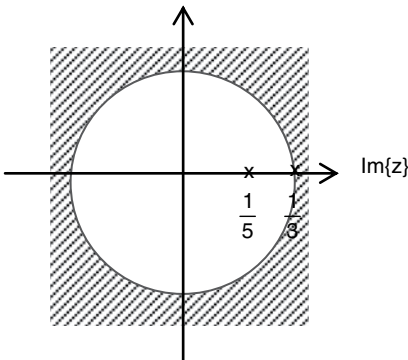
$$X(z) = \frac{z^{-1}\left(3z - \frac{11}{15}\right)}{z^{-2}\left(z - \frac{1}{3}\right)\left(z - \frac{1}{5}\right)} = \frac{1}{z^{-1}} \left[\frac{2}{z - \frac{1}{3}} + \frac{1}{z - \frac{1}{5}} \right]$$

$$X(z) = \frac{2}{1 - \frac{1}{3}z^{-1}} + \frac{1}{1 - \frac{1}{5}z^{-1}}$$

(i) When ROC is $|z| > 1/3$

There are two poles $z = 1/3$ and $1/5$, and ROC lies outside outermost pole, so the inverse z transform is a right-sided sequence $\left(\frac{1}{5}\right)^n u(n) \xleftrightarrow{z} \frac{1}{1 - \frac{1}{5}z^{-1}}, |z| > \frac{1}{5}$

$$\left(\frac{1}{3}\right)^n u(n) \xleftrightarrow{z} \frac{1}{1 - \frac{1}{3}z^{-1}}, |z| > \frac{1}{3}$$



For the above two functions, ROC can be combined as $|z| > \frac{1}{3}$

So, $x(n) = 2\left(\frac{1}{3}\right)^n u(n) + \left(\frac{1}{5}\right)^n u(n)$

(ii) When ROC is

$$\frac{1}{5} < |z| < \frac{1}{3}$$

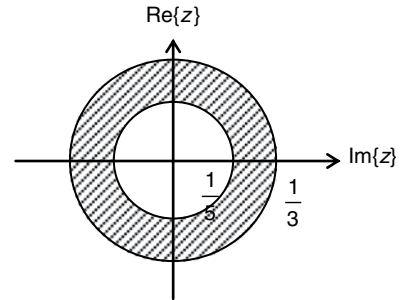
ROC is $|z| < \frac{1}{3}$ and $|z| > \frac{1}{5}$

So the functions

$$\left(\frac{1}{5}\right)^n u(n) \xleftrightarrow{z} \frac{1}{1 - \frac{1}{5}z^{-1}} \quad |z| > \frac{1}{5}$$

$$-\left(\frac{1}{3}\right)^n u[-n-1] \xleftrightarrow{z} \frac{1}{1 - \frac{1}{3}z^{-1}} \quad |z| < \frac{1}{3}$$

By combining above two functions



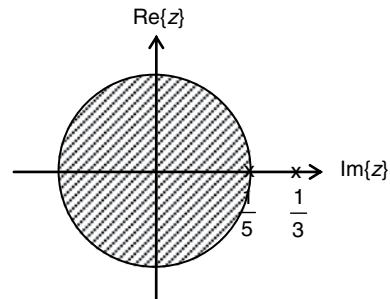
$$x(n) = \left(\frac{1}{5}\right)^n u(n) - 2\left(\frac{1}{3}\right)^n u(-n-1)$$

(iii) When ROC is $|z| < \frac{1}{5}$; in this case, the ROC is inside the both poles

$$-\left(\frac{1}{5}\right)^n u(-n-1) \xleftrightarrow{z} \frac{1}{1 - \frac{1}{5}z^{-1}} \quad |z| < \frac{1}{5}$$

$$-\left(\frac{1}{3}\right)^n u(-n-1) \xleftrightarrow{z} \frac{1}{1 - \frac{1}{3}z^{-1}} \quad |z| < \frac{1}{3}$$

By combining the above two equations



$$x(n) = -2\left(\frac{1}{3}\right)^n u(-n-1) - \left(\frac{1}{5}\right)^n u(-n-1)$$

Example 5

What is the inverse z transform of $X(z) = \log(z + a) - \log z$, $|z| > |a|$

Solution

$$X(z) = \log(z + a) - \log z, |z| > |a|$$

$$= \log\left(\frac{z+a}{z}\right), |z| > |a|$$

$$X(z) = \log(1 + az^{-1}), |z| > |a|$$

By using multiplication by n property,

$$n x(n) \xleftrightarrow{z} -z \frac{dX(z)}{dz} \quad (1)$$

$$\frac{dX(z)}{dz} = \frac{a}{1+az^{-1}} \cdot (-z^{-2})$$

$$-z \frac{dX(z)}{dz} = \frac{az^{-1}}{1+az^{-1}} |z| > |a|$$

$$(-a)^n u(n) \xleftrightarrow{z} \frac{1}{1+az^{-1}}, |z| > |a|$$

$$a(-a)^n u(n) \xleftrightarrow{z} \frac{a}{1+az^{-1}}, |z| > |a|$$

By using time-shifting property

$$a(-a)^{n-1} u[n-1] \xleftrightarrow{z} z^{-1} \frac{a}{1+az^{-1}},$$

$$|z| > |a| \quad (3)$$

So, from equations (1), (2), and (3)

$$nx(n) = a(-a)^{n-1} u(n-1)$$

$$x(n) = \frac{-(-a)^n u(n-1)}{n}$$

Example 6

What is the inverse z transform of $X(z) = 5z^4 + 3z^2 + 2 + z^{-1} + 4z^{-3}$ $0 < |z| < \infty$

Solution

$$\delta(n) \xleftrightarrow{z} 1$$

for any $x(n)$

$$x(n - n_0) \leftrightarrow z^{-n_0} X(z)$$

$$\text{So, } \delta(n+4) \leftrightarrow z^4$$

$$\delta(n+2) \leftrightarrow z^2$$

$$\delta(n-1) \leftrightarrow z^{-1}$$

$$\delta(n-3) \leftrightarrow z^{-3}$$

$$X(z) = 5z^4 + 3z^2 + 2 + z^{-1} + 4z^{-3}$$

Then, $x(n) = 5\delta(n+4) + 3\delta(n+2) + 2\delta(n) + \delta(n-1) + 4\delta(n-3)$

Example 7

Inverse z transform for $X(z) = \frac{z^{-1}}{(1-az^{-1})^2}$, $|z| > |a|$

Solution

$$a^n u(n) \xleftrightarrow{z} \frac{1}{1-az^{-1}} |z| > |a|$$

Consider $x(n) = a^n u(n)$

$$nx(n) \xleftrightarrow{z} -z \frac{d}{dz} \{X(z)\}$$

$$-z \frac{d}{dz} \left\{ \frac{1}{1-az^{-1}} \right\} = -z \frac{-az^{-2}}{(1-az^{-1})^2} = \frac{+az^{-1}}{(1-az^{-1})^2}$$

$$\frac{n}{a} x(n) \xleftrightarrow{z} \frac{z^{-1}}{(1-az^{-1})^2}$$

$$\text{So, the inverse z transform of } \frac{z^{-1}}{(1-az^{-1})^2} \text{ is } \frac{n}{a} x(n) = \frac{n}{a} \cdot a^n u(n) = n \cdot a^{n-1} u(n) \quad (2)$$

ANALYSIS OF DISCRETE-TIME LTI SYSTEMS

Function of a Discrete-Time LTI Systems

The output $y(n)$ of a discrete-time LTI system equals the convolution of the input $x(n)$ with the impulse response $h(n)$ that is $x(n) * h(n)$

By applying z transform $Y(z) = X(z) H(z)$

$$H(z) = \frac{Y(z)}{X(z)}$$

The transform $H(z)$ of $h(n)$ is referred to as the system function (or transfer function) of the system

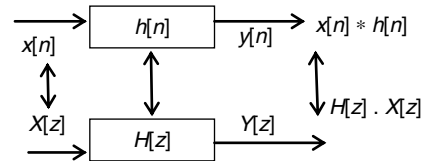


Figure 2 Impulse response and system function.

Characterization of Discrete-Time LTI Systems

(i) Causality

For causal discrete-time LTI system, we have $h(n) = 0, n < 0$.

Since $h(n)$ is right-sided signal, the corresponding requirement on $H(z)$ is that the ROC of $H(z)$ must be of the form $|z| > r_{\max}$, exterior of circle, including infinity.

Similarly, if system is anticausal, ($h(n) = 0, n \geq 0$), the ROC of $H(z)$ must be of the form $|z| < r_{\min}$

(ii) **Stability**

A discrete-time LTI system is BIBO stable if and only if $\sum_{n=-\infty}^{\infty} |h(n)| < \infty$ corresponding requirement on $H(z)$ is that ROC of $H(z)$ contains the unit circle (i.e., $|z| = 1$)

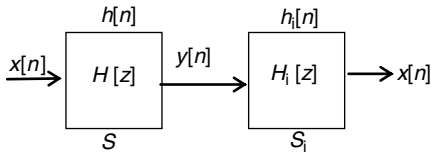
Causal and Stable Systems

If the system is both causal and stable, then all of the poles of $H(z)$ must lie inside the unit circle of the z -plane because the ROC is of the form $|z| > r_{\max}$ and since unit circle is included in ROC, we must have $r_{\max} < 1$

(iii) **Invertibility**

If $H(z)$ is the transfer function of a system S , the S_i , its inverse system has a transfer function $H_i(z) = \frac{1}{H(z)}$.

The inverse system S_i undoes the operation of S . Hence if $H(z)$ is placed in cascade with $H_i(z)$, the transfer function of the composite system is unity.



$H(z) \cdot H_i(z) = 1$, that is, $h(n) \times h_i(n) = \delta(n)$. The zeros of $H(z)$ are the poles of $H_i(z)$, and the poles of $H(z)$ are the zeros of $H_i(z)$ as $H_i(z) = \frac{1}{H(z)}$. If $H(z)$ is both stable and causal, then all poles of $H(z)$ will lie inside the unit circle. If $H_i(z)$ is to be both stable and causal, then all poles of $H_i(z)$ have to lie inside the unit circle, the poles of $H_i(z)$ = zeros of $H(z)$. So, for the systems $H(z)$ and $H_i(z)$ to be stable and causal, all the poles and zero of $H(z)$ must lie inside the unit circle. A system with all poles and zeros inside the unit circle is termed as minimum phase system.

System Function of LTI System Described by Linear Constant Coefficient Difference Equations

A discrete-time LTI system for which input $x(n)$ and output $y(n)$ satisfy the general linear constant coefficient difference equation is of the form

$$\sum_{k=0}^N a_k y(n-k) = \sum_{k=0}^M b_k x(n-k)$$

By applying z transform using time shift property

$$\sum_{k=0}^N a_k z^{-k} Y(z) = \sum_{k=0}^M b_k z^{-k} X(z)$$

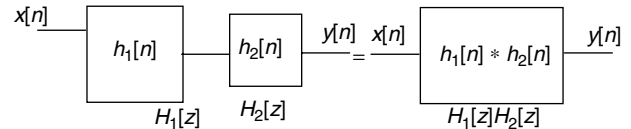
$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}}$$

So, $H(z)$ is always rational

The difference equation by itself does not provide information about which ROC to associate with the algebraic expression $H(z)$. An additional constraint, such as the causality or stability of the system, however, serves to specify the region of convergence. For example, if the system is causal, the ROC will be outside the outermost pole. If the system is stable, the ROC must include the unit circle.

Systems Interconnections

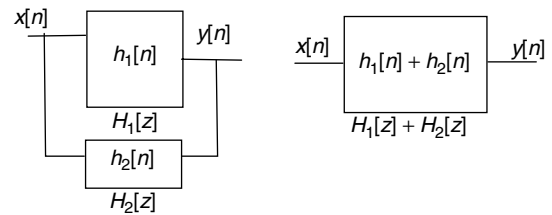
For two LTI system $(h_1(n), h_2(n))$ in cascade, the overall impulse response $h(n)$ is given by $h(n) = h_1(n) \times h_2(n)$ corresponding system function $H(z) = H_1(z) \cdot H_2(z)$



The impulse response of a parallel combination of two LTI systems is given by

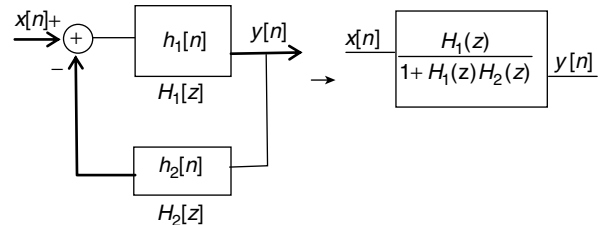
$$h(n) = h_1(n) + h_2(n)$$

$$\text{and } H(z) = H_1(z) + H_2(z)$$



When two systems are connected in feedback interconnection, as shown in figure, then the overall system function for the feedback system

$$H(z) = \frac{Y(z)}{X(z)} = \frac{H_1(z)}{1 + H_1(z)H_2(z)}$$

**Example 8**

Consider the an LTI system for which the input $x(n)$ and output $y(n)$ satisfy the linear constant coefficient difference equations

$$y(n) - \frac{1}{3}y(n-1) = x(n) + \frac{1}{5}x(n-1),$$

then what is the impulse response of the system when
(i) system is stable, (ii) system is anticausal?

Solution

Given difference equations is

$$y(n) - \frac{1}{3}y(n-1) = x(n) + \frac{1}{5}x(n-1)$$

By considering z transform

$$Y(z) - \frac{1}{3}z^{-1}Y(z) = X(z) + \frac{1}{5}z^{-1}X(z)$$

$$Y(z)\left(1 - \frac{1}{3}z^{-1}\right) = X(z)\left(1 + \frac{1}{5}z^{-1}\right)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\left(1 + \frac{1}{5}z^{-1}\right)}{\left(1 - \frac{1}{3}z^{-1}\right)}$$

For ROC, we need to consider the system conditions

(i) When the system is stable, it includes unit circle

$$|z| > \frac{1}{3} \text{ (and the system is causal too)}$$

$$H(z) = \frac{1}{1 - \frac{1}{3}z^{-1}} + \frac{1}{5} \frac{z^{-1}}{1 - \frac{1}{3}z^{-1}} \quad |z| > \frac{1}{3}$$

By considering inverse z transform

$$h(n) = \left(\frac{1}{3}\right)^n u(n) + \frac{1}{5} \left(\frac{1}{3}\right)^{n-1} u(n-1) \text{ is the impulse}$$

response, when the system is causal (or) stable

(ii) When the system is anticausal, $|z| < \frac{1}{3}$

$$H(z) = \frac{1}{1 - \frac{1}{3}z^{-1}} + \frac{1}{5} \frac{z^{-1}}{1 - \frac{1}{3}z^{-1}}, \quad |z| < \frac{1}{3}$$

by considering inverse z transform

$$h(n) = -\left(\frac{1}{3}\right)^n u(-n-1) - \left(\frac{1}{5}\right) \left(\frac{1}{3}\right)^{n-1} u[-(n-1)-1]$$

$$h(n) = -\left(\frac{1}{3}\right)^n u(-n-1) - \left(\frac{1}{5}\right) \left(\frac{1}{3}\right)^{n-1} u(-n)$$

is the impulse response, when the system is anti causal

THE UNILATERAL Z TRANSFORM

The unilateral (or one-sided) z transform $X_1(z)$ of a sequence $x(n)$ is defined as $X_1(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$ and differs from the bilateral transform in that the summation is carried over only $n \geq 0$.

ROC of $X_1(z)$ is always outside a circle in the z plane. Time-shifting property of unilateral z transform if $x(n) \leftrightarrow X_1(z)$, then for $m \geq 0$

$$x(n-m) \leftrightarrow z^{-m} X_1(z) + z^{-m+1} x[-1] + z^{-m+2} x[-2] + \dots + x[-m]$$

$$x(n+m) \leftrightarrow z^m X_1(z) - z^m x[0] - z^{m-1} x[1] - \dots - z x[m-1]$$

The unilateral z transform, that is particularly useful in analysing causal systems specified by linear constant coefficient difference equations with non-zero initial conditions

$$\text{Time delay } x(n-1) \xleftarrow{(\text{unilateral } z)} z^{-1} X(z) + x(-1)$$

$$\text{Time delay } x(n+1) \xleftarrow{(\text{unilateral } z)} z X(z) - z x[0]$$

RELATIONSHIP BETWEEN Z AND LAPLACE TRANSFORM

For a general signal, $x(t)$ is sampled at sampling rate $\frac{1}{T}$ to get discrete value $x(kT)$ which has z transform

$$X(z) = \sum_{k=-\infty}^{\infty} x(kT)z^{-k}$$

The same general signal $x(t)$ can be considered as the impulse sampled at the same rate $\frac{1}{T}$ and may be represented as

$$x(t) = \sum_{k=-\infty}^{\infty} x(kT)\delta(t-kT)$$

Laplace transform of above equation is

$$X(s) = \sum_{k=-\infty}^{\infty} x(kT)e^{-ksT}$$

if $e^{sT} = z$, we can write as

$$X(s) = \sum_{k=-\infty}^{\infty} x(k)z^{-k} = X(z)$$

Thus, $z = e^{sT}$
 $\ln z = sT$.

$$s = \frac{1}{T} \ln z$$

$$X(s) = X(z)|_{z=e^{sT}}$$

STRUCTURES FOR THE REALIZATION OF IIR SYSTEMS

In time domain, the representation N^{th} order IIR system is

$$y(n) = - \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k) \quad (i)$$

and the z-domain representation of an N^{th} order IIR system is,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

In the above representation, the value of M gives the number of zeros and N gives the number of poles of the system.

Types of structures for realizing the IIR systems are as follows:

Direct Form – I Structure

Consider the difference equation governing an IIR system.
Consider the equation (1)

$$y(n) + a_1 y(n-1) + a_2 y(n-2) + \dots + a_N y(n-N) = b_0 x(n) + b_1 x(n-1) + b_2 x(n-2) + \dots + b_M x(n-M)$$

on taking z transform both sides
we get,

$$Y(z) = -a_1 z^{-1} Y(z) - a_2 z^{-2} Y(z) - \dots - a_N z^{-N} Y(z) + b_0 X(z) + b_1 z^{-1} X(z) + \dots + b_M z^{-M} X(z)$$

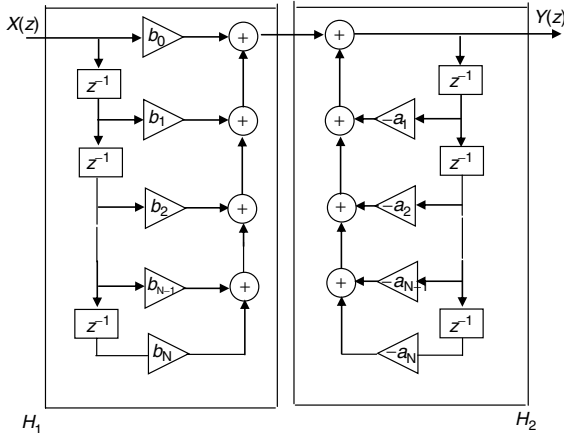


Figure 3 Direct form-I structure of IIR system.

This provides a direct relation between time-domain and z-domain equations.

⇒ This required more memory elements (z^{-1}).

Direct Form-II structure of IIR system

This structure can be realized which uses less number of delay elements or memory elements than the direct form-I. Consider the general difference equation governing an IIR system.

$$Y(n) = - \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

on taking z transform of the above equation.

We can obtained

$$Y(z) [1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_N z^{-N}] = X(z) [b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_M z^{-M}]$$

$$\frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

$$\text{Let } \frac{Y(z)}{X(z)} = \frac{W(z)}{X(z)} \frac{Y(z)}{W(z)}$$

$$\text{Where, } \frac{W(z)}{X(z)} = \frac{1}{1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_N z^{-N}} \quad (2)$$

$$\frac{Y(z)}{W(z)} = b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_M z^{-M} \quad (3)$$

On cross multiplying equation (2) and (3), we get

$$W(z) = X(z) - a_1 z^{-1} W(z) - a_2 z^{-2} W(z) - \dots - a_N z^{-N} W(z) \quad (4)$$

and

$$Y(z) = b_0 W(z) + b_1 z^{-1} W(z) + \dots + b_M z^{-M} W(z) \quad (5)$$

The equations (4) and (5) represent the IIR system in z-domain and it can be realized by a direct structure called direct form-II.

In direct form-II, the number of delays is equal to order of the system.

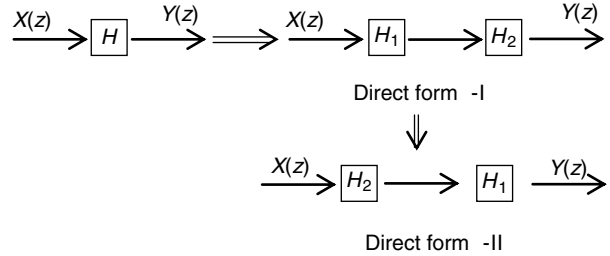


Figure 4 Conversion of Direct form-I to Direct form-II.

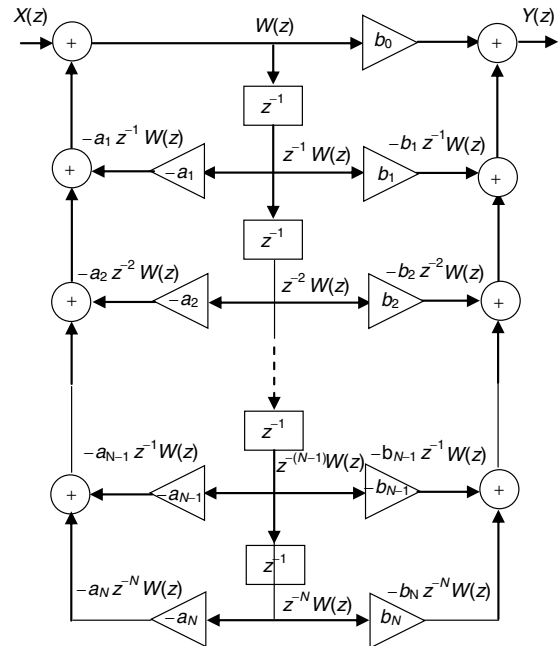
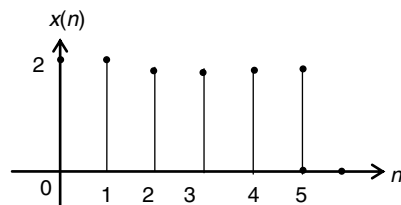


Figure 5 Direct form-II structure of IIR system.

Solved Examples

Example 9

What is the z transform of the signal shown in figure?



Solution

Here, $x(0) = x(1) = \dots = x(5) = 2$, and $x(6) = x(7) = \dots = 0$

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} x(n)z^{-n} \\ &= 2 + \frac{2}{z} + \frac{2}{z^2} + \frac{2}{z^3} + \frac{2}{z^4} + \frac{2}{z^5} \\ &= \frac{2}{z^5} (z^5 + z^4 + z^3 + z^2 + z + 1) \text{ for all } z \neq 0 \end{aligned}$$

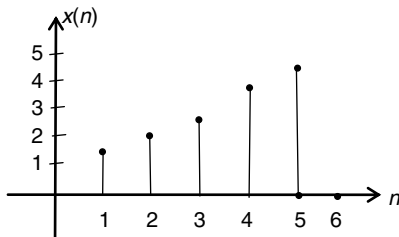
$$\text{(or)} \quad X(z) = 2 \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{z^4} + \frac{1}{z^5} \right)$$

This expression in geometric progression, the summation of n terms with initial value a is $S_n = \frac{a(r^n - 1)}{r - 1}$

$$\text{So,} \quad X(z) = 2 \left(\frac{1 \left(\left(\frac{1}{z} \right)^6 - 1 \right)}{\left(\frac{1}{z} - 1 \right)} \right) = \frac{2z}{z-1} (1 - z^{-6})$$

Example 10

Find the z transform of the signal $x(n]$ depicted in the figure

**Solution**

The signal $x(n]$ can be expressed as

$$x(n) = n \text{ for } 0 \leq n \leq 5$$

$$x(n) = n \{u(n) - u(n-6)\}$$

$$= n u(n) - n u(n-6)$$

$$= n u(n) - (n-6+6) u(n-6)$$

$$= n u(n) - (n-6) u(n-6) - 6 u(n-6)$$

$$u(n) \leftrightarrow \frac{1}{1-z^{-1}}$$

$$n u(n) \leftrightarrow \frac{z^{-1}}{(1-z^{-1})^2}$$

$$x(n-k) \leftrightarrow z^{-k} X(z)$$

So applying z transform

$$X(z) = \frac{z^{-1}}{(1-z^{-1})^2} - \frac{1}{z^6} \frac{z^{-1}}{(1-z^{-1})^2} - \frac{1}{z^6} \frac{6}{(1-z^{-1})}, |z| > 1$$

$$\begin{aligned} X(z) &= \frac{z^{-1} - z^{-7} - 6z^{-6}(1-z^{-1})}{(1-z^{-1})^2} = \frac{5z^{-7} - 6z^{-6} + z^{-1}}{(1-z^{-1})^2} \\ &= \frac{z^6 - 6z + 5}{z^5(z-1)^2} \end{aligned}$$

Example 11

A discrete LTI system has the difference equation $y(n+1] - 2y(n) = x(n+1]$, if the initial conditions are $y(-1) = -1$, and

the input $x(n) = \left(\frac{1}{3}\right)^n u(n)$ then find output $y(n)$.

Solution

Given $y(n+1] - 2y(n) = x(n+1]$

$$y(n) - 2y(n-1) = x(n)$$

$$y(n) u(n) \leftrightarrow Y(z)$$

$$y(n-1] u(n) \leftrightarrow \frac{1}{z} Y(z) + y[-1] = \frac{1}{z} Y(z) + (-1)$$

$x(-1) = x(-2) = \dots = x(-n) = 0$, $x(n)$ is causal input

$$x(n) = \left(\frac{1}{3}\right)^n u(n)$$

$$\Rightarrow \quad X(z) = \frac{1}{1 - \frac{1}{3}z^{-1}}$$

By taking the z transform for the difference equation

$$Y(z) - 2 \left[\frac{1}{2} Y(z) - 1 \right] = X(z)$$

$$Y(z) \left[1 - \frac{2}{z} \right] + 2 = X(z)$$

$$Y(z) = \frac{X(z) - 2}{1 - 2z^{-1}} = \frac{\frac{1}{1 - \frac{1}{3}z^{-1}} - 2}{1 - 2z^{-1}}$$

$$= \frac{1 - 2 + \frac{2}{3}z^{-1}}{(1 - 2z^{-1}) \left(1 - \frac{1}{3}z^{-1} \right)} = \frac{-1 + \frac{2}{3}z^{-1}}{(1 - 2z^{-1}) \left(1 - \frac{1}{3}z^{-1} \right)}$$

$$Y(z) = \frac{z(-3z+2)}{(z-2)(3z-1)} = z \left[\frac{A}{z-2} + \frac{B}{3z-1} \right]$$

$$= z \left[\frac{-4}{5} \frac{1}{z-2} - \frac{3}{5} \frac{1}{(3z-1)} \right] = \frac{-1}{5} \left[\frac{4}{1-2z^{-1}} + \frac{1}{1-\frac{1}{3}z^{-1}} \right]$$

$$\begin{aligned} \Rightarrow \quad y(n) &= \frac{-1}{5} \left[4.2^n u(n) + \left(\frac{1}{3} \right)^n u(n) \right] \\ &= -0.2 \left[2^{n+2} + \left(\frac{1}{3} \right)^n \right] u(n) \end{aligned}$$

Example 12

For a causal system specified by the transfer function $H(z) = \frac{z}{z-0.4}$, find the zero-state response to input

$$x(n) = (0.5)^n u(n) + 2 \cdot 2^n u(-n-1)$$

Solution

Zero-state response means $y(n)$ with zero initial conditions

$$\begin{aligned} X(z) &= \frac{1}{1-0.5z^{-1}} - \frac{2}{1-2z^{-1}} = \frac{z(z-2-2z+1)}{(z-0.5)(z-2)} \\ &= \frac{-z(z+1)}{(z-0.5)(z-2)} \end{aligned}$$

The ROC of $X(z)$ is

For the first term $|z| > 0.5$ and For the second term $|z| < 2$

The ROC for $X(z)$ is common region given by $0.5 < |z| < 2$

$$\text{and } X(z) = \frac{-z(z+1)}{(z-0.5)(z-2)}$$

$$\text{Then } Y(z) = X(z) \cdot H(z) = \frac{-z^2(z+1)}{(z-0.4)(z-0.5)(z-2)}$$

as $H(z)$ is causal system ROC is $|z| > 0.4$

The common region of convergence of $X(z)$ and $H(z)$ is $0.5 < |z| < 2$, so

$$Y(z) = \frac{-z^2(z+1)}{(z-0.4)(z-0.5)(z-2)}$$

$$\begin{aligned} \frac{Y(z)}{z} &= \frac{-z(z+1)}{(z-0.4)(z-0.5)(z-2)} \\ &= \frac{-7}{2} \frac{1}{(z-0.4)} + \frac{5}{(z-0.5)} - \frac{5}{2} \frac{1}{z-2} \end{aligned}$$

$$Y(z) = \frac{-7}{2} \frac{z}{z-0.4} + 5 \cdot \frac{z}{z-0.5} - \frac{5}{2} \frac{z}{z-2}, \quad 0.5 < |z| < 2$$

$$y(n) = \left[\frac{-7}{2} (0.4)^n + 5(0.5)^n \right] u(n) + \frac{5}{2} 2^n u(-n-1)$$

Example 13

Find the z transform of the signal $x(n)$

$$= \left\{ n \left(\frac{-1}{3} \right)^n u(n) \right\} * (5)^n u(-n)$$

Solution

We can write $x(n) = w(n) \times y(n)$

$$w(n) = n \left(\frac{-1}{3} \right)^n u(n)$$

$$\left(\frac{-1}{3} \right)^n u(n) \xrightarrow{z} \frac{1}{1+\frac{1}{3}z^{-1}} \text{ with ROC } |z| > \frac{1}{3}$$

$$w(n) = n \left(\frac{-1}{3} \right)^n u(n) \xrightarrow{z} W(z)$$

$$= -z \frac{d}{dz} \left\{ \frac{1}{1+\frac{1}{3}z^{-1}} \right\}$$

$$\begin{aligned} W(z) &= -z \frac{-1}{\left(1+\frac{1}{3}z^{-1}\right)^2} \left(\frac{-1}{3} z^{-2} \right) \\ &= \frac{-1}{3} \frac{z^{-1}}{\left(1+\frac{1}{3}z^{-1}\right)^2} \text{ with ROC } |z| > \frac{1}{3} \end{aligned}$$

$$y(n) = (5)^n u(-n) = \left(\frac{1}{5} \right)^{-n} u[-n]$$

$$\left(\frac{1}{5} \right)^n u(n) \xleftrightarrow{z} \frac{z}{z-\frac{1}{5}}, \text{ ROC } |z| > \frac{1}{5}$$

$$x(-n) \xleftrightarrow{z} X\left(\frac{1}{z}\right) \text{ with ROC } \frac{1}{R_x}$$

$$\begin{aligned} \text{So, } Y(z) &= \frac{\frac{1}{z}}{\frac{1}{z}-\frac{1}{5}} \text{ ROC } \frac{1}{|z|} > \frac{1}{5} \\ &= \frac{-z^{-1}}{\frac{1}{5}-z^{-1}} \quad |z| < 5 \end{aligned}$$

By applying convolution $y(n) \times w(n)$, and z transform, we have $X(z) = Y(z) W(z)$

$$= \frac{-z^{-1}}{\frac{1}{5}-z^{-1}} \cdot \frac{-1}{3} \frac{z^{-1}}{\left(1+\frac{1}{3}z^{-1}\right)^2}$$

$$\begin{aligned} \text{the common ROC is } |z| > \frac{1}{3}, \text{ and } |z| < 5 \text{ and } X(z) &= \frac{5}{z-5} \cdot \frac{1}{3} \frac{z}{\left(z+\frac{1}{3}\right)^2} \\ &= \frac{5}{3} \frac{z}{(z-5)\left(z+\frac{1}{3}\right)^2} \text{ with ROC } \frac{1}{3} < |z| < 5 \end{aligned}$$

Example 14

Find the transfer function and impulse response of a causal LTI system, if the input to the system is $x(n) = u(n)$ and output is $y(n) = \delta(n)$

Solution

The z transform of inputs and outputs

$$X(z) = \frac{1}{1-z^{-1}}, \text{ ROC } |z| > 1,$$

$$Y(z) = 1, \text{ ROC for all } z$$

Then transfer function $H(z)$

$$= \frac{Y(z)}{X(z)} = \frac{1}{1/(1-z^{-1})}$$

$= 1 - z^{-1}$, with ROC all z except at $z = 0$

By taking inverse z transform,

$h(n) = \delta(n) - \delta(n - 1)$ is the impulse response.

Example 15

An LTI system is described by the difference equation

$$y(n) - \frac{1}{2} y(n-1) + \frac{1}{18} y(n-2) = x(n) + \frac{1}{20} x(n-1) - \frac{1}{20} x(n-2)$$

Does a stable and causal LTI inverse system exists?

Solution

$$y(n) - \frac{1}{2} y(n-1) + \frac{1}{18} y(n-2) = x(n) + \frac{1}{20} x(n-1) - \frac{1}{20} x(n-2)$$

By considering z transform

$$Y(z) - \frac{1}{2} z^{-1} Y(z) + \frac{1}{18} z^{-2} Y(z) = X(z) +$$

$$\frac{z^{-1}}{20} X(z) - \frac{1}{20} z^{-2} X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + \frac{1}{20} z^{-1} - \frac{1}{20} z^{-2}}{1 - \frac{1}{2} z^{-1} + \frac{1}{18} z^{-2}} = \frac{z^2 + \frac{1}{20} z - \frac{1}{20}}{z^2 - \frac{1}{2} z + \frac{1}{18}}$$

$$= \frac{\left(z + \frac{1}{4}\right)\left(z - \frac{1}{5}\right)}{\left(z - \frac{1}{3}\right)\left(z - \frac{1}{6}\right)} = \frac{\left(z + \frac{1}{4}\right)\left(z - \frac{1}{5}\right)}{\left(z - \frac{1}{3}\right)\left(z - \frac{1}{6}\right)}$$

$$H_i(z) = \frac{1}{H(z)} = \frac{\left(z - \frac{1}{3}\right)\left(z - \frac{1}{6}\right)}{\left(z + \frac{1}{4}\right)\left(z - \frac{1}{4}\right)}$$

is the transfer function of inverse system. All the poles and zeros of $H(z)$ lie inside the unit circle, so stable and causal inverse system exists.

And this is a minimum phase system.

Example 16

A discrete time LTI model for two path communication channel is $y(n) = x(n) + a x(n-1)$, Find the difference equation description of the inverse system.

Solution

$y(n) = x(n) + ax(n-1)$, By taking z transform $Y(z) = X(z) + az^{-1} X(z)$

$$\text{Transfer function } H(z) = \frac{Y(z)}{X(z)} = 1 + a z^{-1}$$

The inverse system has transfer function

$$H_i(z) = \frac{1}{H(z)} = \frac{1}{1 + a z^{-1}} = \frac{Y_i(z)}{X_i(z)}$$

$$Y_i(z) + a z^{-1} Y_i(z) = X_i(z),$$

By taking inverse z transform

$$y(n) + a y(n-1) = x(n)$$

The inverse system is both stable and causal if $|a| < 1$.

EXERCISES

Practice Problems 1

Direction for questions 1 to 22: Select the correct alternative from the given choices.

1. $x(n) = a^n u(n)$ where a is real. Z transform of $x(n)$ is

(A) $\frac{z}{z-a}$, $|z| < |a|$ (B) $\frac{z}{z-a}$, $|z| > |a|$

(C) $\frac{z}{z+a}$, $|z| > |a|$ (D) $\frac{z}{z+a}$, $|z| < |a|$

2. Z transform of $a^{n+1} u(n+1)$ is

(A) $\frac{z^2}{z-a}$, $|a| < |z| < \infty$ (B) $\frac{z^2}{z+a}$, $|a| < |z| < \infty$

(C) $\frac{z}{z-a}$, $|a| < |z| < \infty$ (D) $\frac{z^2}{z-a}$, $|a| > |z| < \infty$

3. Z transform of $na^n u(n)$ is

(A) $\frac{-az}{(z-a)^2}$, $|z| > |a|$ (B) $\frac{az}{(z-a)^2}$, $|z| > |a|$

(C) $\frac{az}{(z-a)^2}$, $|z| > |a|$ (D) $\frac{az}{(z-a)^2}$, $|z| < |a|$

4. Inverse z transform of $\log(1 - az^{-1})$, $|z| > |a|$ is

(A) $\frac{1}{n} a^n u(n-1)$ (B) $\frac{-1}{n} a^n u(n-1)$

(C) $\frac{-1}{n} a^n u(n+1)$ (D) $\frac{1}{n} a^n u(n+1)$

5. Find the initial and final values of z -domain signal

$$\frac{2z^{-1}}{1 - 1.8z^{-1} + 0.8z^{-2}}$$

(A) 10, 1 (B) 1, 10 (C) 0, 10 (D) 0, ∞

6. The transfer function of a system is $H(z) = 1 - 3z^{-1}$, the system response for input $x(n)$ is
 (A) $x(n) - 0.3 x(n-1)$
 (B) $x(n) + 0.3 x(n-1)$
 (C) $x(n) - 3 x(n+1)$
 (D) $x(n) - 3 x(n-1)$

7. Unilateral z transform of $\left(\frac{1}{3}\right)^n u(n) + \left(\frac{1}{5}\right)^n u(n)$ and ROC are respectively?

- (A) $\frac{z}{(z-1/3)} + \frac{z}{(z-1/5)}; |z| > \frac{1}{5}$
 (B) $\frac{z}{(z-1/3)} + \frac{z}{(z-1/5)}; |z| > \frac{1}{3}$
 (C) $\frac{z}{(z-1/3)} + \frac{z}{(z-1/3)}; \frac{1}{5} > |z| > \frac{1}{3}$
 (D) $\frac{z}{(z-1/3)} + \frac{z}{(z-1/3)}; \frac{1}{5} < |z| < \frac{1}{3}$

8. Find $f(0), f(1), f(2)$, and $f(3)$, if

$$F(z) = \frac{z^2 + 2}{z^3 - 3z^2 + 3z - 1}$$

- (A) 0, 9, 31, 16
 (B) 0, 1, 3, 8
 (C) 1, 3, 3, 1
 (D) 0, 1, 2, 4

9. The minimum Nyquist sampling rate $\frac{\sin t}{\pi t}$ is

- (A) $2/\pi$ (B) $1/\pi$ (C) π (D) 2π

10. Match the following:

List-I	List-II
$x(n)$	ROC
(1) $\delta(n)$	(a) All z
(2) $u(n)$	(b) $ z > 1$
(3) $-u(-n-1)$	(c) All z except 0
(4) $\delta(n-m)$	(d) $ z < 1$

- 1 2 3 4
 (A) a b c d
 (C) a b d c
- 1 2 3 4
 (B) b a c d
 (D) b a d c

11. $y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n)$ where $x(n)$ and $y(n)$ are the input and output of the systems, then determine system function.

- (A) $\frac{z^2}{\left(z + \frac{1}{2}\right)\left(z - \frac{1}{4}\right)}; |z| > \frac{1}{2}$
 (B) $\frac{z^2}{\left(z + \frac{1}{2}\right)\left(z - \frac{1}{4}\right)}; |z| < \frac{1}{2}$

- (C) $\frac{z^2}{\left(z + \frac{1}{2}\right)\left(z - \frac{1}{4}\right)}; |z| > \frac{1}{2}$
 (D) $\frac{z^2}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{4}\right)}; |z| > \frac{1}{2}$

12. For a Z transform $X(z) = \frac{z(2z-5/6)}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{3}\right)}$. Find out the

sequences when the region of convergence is $|z| > 1/2$

- (A) $\left[\left(\frac{1}{2}\right)^n - \left(\frac{1}{3}\right)^n u(n)\right]$
 (B) $\left(\frac{1}{2}\right)^n u(n) - \left(\frac{1}{3}\right)^n u(-n-1)$
 (C) $\left[\left(\frac{1}{2}\right)^n + \left(\frac{1}{3}\right)^n\right] u(n)$
 (D) $\left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{3}\right)^n u(-n-1)$

13. Find the sequence when region of convergence

$\frac{1}{3} < |z| < \frac{1}{2}$ for above $X(z)$,

- (A) $-\left(\frac{1}{2}\right)^n u(-n-1) + \left(\frac{1}{3}\right)^n u(n)$
 (B) $\left(\frac{1}{2}\right)^n u(-n-1) + \left(\frac{1}{3}\right)^n u(n)$
 (C) $\left(\frac{1}{2}\right)^n u(n) - \left(\frac{1}{3}\right)^n u(-n-1)$
 (D) $\left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{3}\right)^n u(-n-1)$

14. Impulse response for the cascade of LTI systems hav-

ing impulse response $h_1(n) = \left(\frac{1}{2}\right)^n u(n)$ and $h_2(n) = \left(\frac{1}{4}\right)^n u(n)$ is?

- (A) $\left(\frac{1}{4}\right)^n (2n-1)u(n)$ (B) $\left(\frac{1}{4}\right)^n (2^{n+1}-1)u(n)$
 (C) $\left(\frac{1}{4}\right)^n (2^{n+1}+1)u(n)$ (D) $\left(\frac{1}{4}\right)^n (2^n+1)u(n)$

15. What is the Nyquist rate for the signal $x(t) = \cos 2000\pi t + 6\sin 9000\pi t$

- (A) 9 kHz (B) 4 kHz
 (C) 2 kHz (D) 11 kHz

16. Which of the following is the inverse Z transform of

$$X(z) = \frac{z}{(z-2)(z-3)} \quad |z| < 2$$

- (A) $[2^n - 3^n] u(-n-1)$
 (B) $[2^n - 3^n] u(n+1)$
 (C) $[3^n - 2^n] u(-n-1)$
 (D) $[2^n - 3^n] u(n)$

17. The impulse response of a relaxed linear time invariant system is $h(n) = \alpha^n u(n)$ with $|\alpha| < 1$. Determine the value of the step response as $n \rightarrow \infty$

- (A) $\left[\frac{1-\alpha^{n+1}}{1-\alpha} \right] u(n)$ (B) $\left[\frac{1-\alpha^{n-1}}{1-\alpha} \right] u(n)$
 (C) $\left[\frac{1+\alpha^{n+1}}{1-\alpha} \right] u(n)$ (D) $\left[\frac{1+\alpha^{n+1}}{1-\alpha} \right] u(n)$

18. Consider Z transform $X(z) = 5z^2 + 4z^{-1} + 3; 0 < |z| < \infty$, the inverse z transform $x(n)$ is?

- (A) $5\delta(n+1)$
 (B) $3\delta(n)$
 (C) $4\delta(n+2)$
 (D) $5\delta(n+2) + 4\delta(n-1) + 3\delta(n)$

19. The transfer function of D.T, LTI system is given by

$$H(z) = \frac{2 - \frac{3}{4}z^{-1}}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}}$$

Consider the following statements

S1: System is stable and causal for ROC $|z| > \frac{1}{2}$

S2: System is stable but not causal for ROC: $|z| < \frac{1}{4}$

S3: System is neither stable nor causal for ROC:

$$\frac{1}{4} < |z| < \frac{1}{2}$$

Which one of the following statements is valid?

- (A) Both S1 and S2 are true
 (B) Both S2 and S3 are true
 (C) Both S1 and S3 are true
 (D) All are true

20. ROC of z transform of the discrete time sequence $x(n)$

$$= \left(\frac{1}{3} \right)^n u(n) - \left(\frac{1}{2} \right)^n u[-n-1]$$
 is

- (A) $|z| > \frac{1}{3}$ (B) $|z| < \frac{1}{2}$
 (C) $|z|$ for all (D) $\frac{1}{3} < |z| < \frac{1}{2}$

21. A causal LTI system is described by the difference equation $2y(n) = \alpha y(n-1) - 2x(n) + \beta x(n-1)$. The system is stable for which values of α, β ?

- (A) $|\alpha| > 2, |\beta| < 1$ (B) $|\alpha| < 2$, any β
 (C) $|\alpha| < 3, |\beta| > 1$ (D) $|\alpha| > 2$, any β

22. A sequence $x(n]$ with z transform $X(z) = z^4 + z^2 - 2z + 2 - 3z^{-4}$ is applied as an input to a linear, time-invariant system with impulse response $h(n) = 2\delta(n-3)$, the output at $n = 4$ is

- (A) 2 (B) 0 (C) -2 (D) -6

Practice Problems 2

Direction for questions 1 to 21: Select the correct alternative from the given choices.

1. Two D.T systems with impulse response $h_1(n) = \delta(n-1)$, $h_2(n) = \delta(n-2)$ are connected in cascade, the overall Impulse response of the cascaded system is?

- (A) $\delta(n-1)$ (B) $\delta(n-2)$
 (C) $\delta(n-3)$ (D) $\delta(n-4)$

2. A system with transfer function $H(z)$ has impulse response $h(n)$, defined as $h(2) = 1$, $h(3) = -1$, and $h(n) = 0$ otherwise, consider following statements

S1: $H(z)$ is a low-pass filter

S2: $H(z)$ is an FIR filter

Which one of the following is correct?

- (A) Only S2 is true
 (B) Both S1 and S2 are false
 (C) Both S1 and S2 are true, and S1 is reason for S2
 (D) Both S1 and S2 are true, and S2 is a reason for S1

3. Z transform $X(Z)$ of a sequence $x(n)$ is given by $X(Z) =$

$$\frac{0.5}{1-2z^{-1}}, \text{ it is given that the ROC of } X(Z) \text{ includes unit circle, the value of } x(0) \text{ is}$$

- (A) 0 (B) 0.5 (C) 1 (D) ∞

4. If ROC of $x_1(n) + x_2(n)$ is $\frac{1}{3} < |z| < \frac{2}{3}$, then the ROC of $x_1(n) - x_2(n)$ includes

- (A) $\frac{1}{3} < |z| < 3$ (B) $\frac{2}{3} < |z| < 3$
 (C) $\frac{3}{2} < |z| < 3$ (D) $\frac{1}{3} < |z| < \frac{2}{3}$

5. $y(n) = \sum_{k=0}^n x(k)$

Which one of the following correctly relate the z transform of the input and output denoted by $X(z)$ and $Y(z)$, respectively?

- (A) $Y(z) = \frac{d}{dz} X(z)$ (B) $Y(z) = \frac{X(z)}{1-z^{-1}}$
 (C) $Y(z) = X(z)z^{-1}$ (D) $Y(z) = (1-z^{-1})X(z)$
6. $x(n) = \frac{1}{7} \left(\frac{1}{2} \right)^n u(n) - \frac{6}{7} (3)^n u(-n-1)$. Find out ROC of Z transform?
 (A) $1/2 > |z| < 3$ (B) $|z| > 1/2$
 (C) $|z| < 3$ (D) $1/2 < |z| < 3$
7. The range of values a and b for which the z transform with impulse response $h(n) = a^n, n \geq 0 = b^n, n < 0$ will be stable
 (A) $|a| < 1, |b| < 1$ (B) $|a| < 1, |b| > 1$
 (C) $|a| > 1, |b| > 1$ (D) $|a| > 1, |b| < 1$
8. The Z transform of $a^{-n} u(-n-1)$ is
 (A) $\frac{az}{1+az}, |z| < \frac{1}{|a|}$ (B) $\frac{az}{1-az}, |z| < \frac{1}{|a|}$
 (C) $\frac{-az}{1-az}, |z| < \frac{1}{|a|}$ (D) $\frac{az}{1-az}, |z| > \frac{1}{|a|}$
9. Z transform of $x(n) = \{3, -2, 4, 1, 0, -3\}$ is
 (A) $3z^2 + 2z + 4 + z^{-1} + 3z^{-2}, 2 < |z| < 4$
 (B) Does not exist
 (C) $3z^2 - 2z + 4 + z^{-1} - 3z^{-3}, 0 < |z| < \infty$
 (D) $3z^2 - 2z + 4 + z^{-1} - 3z^{-3}, |z| < 0, |z| < \infty$
10. Z transform of $u(n+1)$ is
 (A) $\frac{z}{z-1}$ (B) $\frac{z^2}{z-1}$
 (C) $\frac{z^{-1}}{1-z^{-1}}$ (D) $\frac{1}{z^{-1}+z^{-2}}$
11. Inverse z transform of $\frac{2}{z-2}, (|z| > 2)$ is
 (A) $2^{n-1} u(n+1)$ (B) $2^{n+1} u(n-1)$
 (C) $2^n u(n+1)$ (D) $2^n u(n-1)$
12. Z transform and Laplace transform are related by
 (A) $s = \ln z$ (B) $s = \frac{\ln z}{T}$
 (C) $s = T \ln z$ (D) Not related
13. The final value of $x(n)$ is
 (A) $\lim_{z \rightarrow \infty} Lt \ zX(z)$ (B) $\lim_{z \rightarrow 0} Lt \ zX(z)$
 (C) $\lim_{z \rightarrow \infty} Lt \ (1-z^{-1})X(z)$ (D) $\lim_{z \rightarrow 1} Lt \ (1-z^{-1})X(z)$
14. The initial value of $x(n)$, if $X(z) = \frac{3z^2}{(z+3)(z-3)}$
 (A) 3 (B) 6
 (C) ∞ (D) 0
15. An LTI system defined as $y(n] + 0.5y(n-1) = x(n] + 0.3x(n-1)$. The transfer function in z plane is
 (A) $\frac{z+0.3}{z+0.5}$ (B) $\frac{z-0.3}{z-0.5}$
 (C) $\frac{z+0.3}{z-0.5}$ (D) $\frac{z-0.3}{z+0.5}$
16. A signal $x(n] = \{1, -1, 2, 1, 3, 2\}$. The ROC z transform $X(z)$ is
 (A) $|z| > 3$
 (B) $-1 < |z| < 3$
 (C) $|z| < 3$
 (D) the entire z plane except at $z = 0$
17. The minimum sampling frequency of $x(t) = 5 \cos 50\pi t + 2 \cos 200\pi t - 10 \cos 100\pi t$ is
 (A) 100 Hz (B) 200 Hz
 (C) 50 Hz (D) Cannot be sampled
18. The minimum number of samples required to exactly describe $x(t) = 10 \cos 4\pi t + 4 \sin 8\pi t$
 (A) 4 samples per second.
 (B) 2 samples per second.
 (C) 8 samples per second.
 (D) 16 samples per second.
19. Aliasing occurs when sampling frequency ω_s is (ω_m – band limited signal frequency)
 (A) $\omega_s = 2\omega_m$ (B) $\omega_s < 2\omega_m$
 (C) $\omega_s > 2\omega_m$ (D) 0
20. Interpolation is the process of
 (A) inserting $N-1$ unity sequence values to $x(n)$
 (B) inserting $N-1$ zero sequence values to $x(n)$
 (C) deleting $N-1$ unity sequence values from $x(n)$
 (D) deleting $N-1$ zero sequence values from $x(n)$
21. Given signal $m(t) = \sin 5\pi t + 5 \sin 10\pi t$ is sampled instantaneously, the maximum interval between samples is
 (A) 0.1 seconds. (B) 0.2 seconds.
 (C) 10 seconds. (D) none.

PREVIOUS YEARS' QUESTIONS

1. The z transform of a system is

$$H(z) = \frac{z}{z-0.2}$$

If the ROC is $|z| < 0.2$, then the impulse response of the system is [2004]

- (A) $(0.2)^n u(n)$ (B) $(0.2)^n u(-n-1)$
 (C) $-(0.2)^n u(n)$ (D) $-(0.2)^n u(-n-1)$

2. A 1 kHz sinusoidal signal is ideally sampled at 1,500 samples/sec and the sampled signal is passed through an ideal low-pass filter with cut-off frequency 800 Hz. The output signal has the frequency? [2004]
 (A) Zero Hz (B) 0.75 kHz
 (C) 0.5 kHz (D) 0.25 kHz
3. A casual LTI system is described by the difference equation

$$2y(n) = \alpha y(n-2) - 2x(n) + \beta x(n-1).$$

The system is stable only if [2004]

- (A) $|\alpha| = 2, |\beta| < 2$
 (B) $|\alpha| > 2, |\beta| > 2$
 (C) $|\alpha| < 2$, any value of β
 (D) $|\beta| < 2$, any value of α

4. The region of convergence of z transform of the sequence $\left(\frac{5}{6}\right)^n u(n) - \left(\frac{6}{5}\right)^n u(-n-1)$ must be [2005]

- (A) $|z| < \frac{5}{6}$ (B) $|z| > \frac{6}{5}$
 (C) $\frac{5}{6} < |z| < \frac{6}{5}$ (D) $\frac{6}{5} < |z| < \infty$

5. If the region of convergence of $X_1(n) + X_2(n)$ is $\frac{1}{3} < |z| < \frac{2}{3}$, then the region of convergence of $X_1(n) - X_2(n)$ includes [2006]

- (A) $\frac{1}{3} < |z| < 3$ (B) $\frac{2}{3} < |z| < 3$
 (C) $\frac{3}{2} < |z| < 3$ (D) $\frac{1}{3} < |z| < \frac{2}{3}$

6. The minimum sampling frequency (in samples/sec) required to reconstruct the following signal from its samples without distortion

$$x(t) = 5 \left(\frac{\sin 2\pi 1000t}{\pi t} \right)^3 + 7 \left(\frac{\sin 2\pi 1000t}{\pi t} \right)^2$$

$$x(t) = 5 \left(\frac{\sin 2\pi 1000t}{\pi t} \right)^3 + 7 \left(\frac{\sin 2\pi 1000t}{\pi t} \right)^2 \quad \text{would be:} \quad [2006]$$

- (A) 2×10^3 (B) 4×10^3
 (C) 6×10^3 (D) 8×10^3

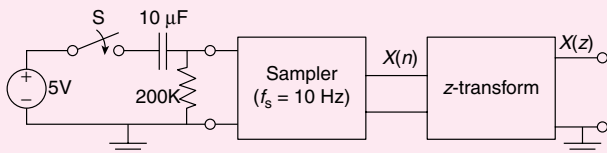
7. The z transform $X(z)$ of a sequence $x(n)$ is given

$$X(z) = \frac{0.5}{1 - 2Z^{-1}}. \text{ It is given that the region of convergence of } X(z) \text{ includes the unit circle. The value of } x(0) \text{ is:} \quad [2007]$$

- (A) -0.5 (B) 0 (C) 0.25 (D) 0.5

Direction for questions 8 and 9:

In the following network, the switch is closed at $t = 0^-$ and the sampling starts from $t = 0$. The sampling frequency is 10 Hz.



8. The samples $x(n]$, ($n = 0, 1, 2, \dots$) are given by [2008]

- (A) $5(1 - e^{-0.05n})$ (B) $5e^{-0.05n}$
 (C) $5(1 - e^{-5n})$ (D) $5e^{-5n}$

9. The expression and the region of convergence of the z transform of the sampled signal are [2008]

- (A) $\frac{5z}{z - e^{-5}}, |z| < e^{-5}$ (B) $\frac{5z}{z - e^{-0.05}}, |z| < e^{-0.5}$
 (C) $\frac{5z}{z - e^{-0.05}}, |z| > e^{-0.05}$ (D) $\frac{5z}{z - e^{-5}}, |z| > e^{-5}$

10. The ROC of Z transform of the discrete time sequence

$$x(n) = \left(\frac{1}{3}\right)^n u(n) - \left(\frac{1}{2}\right)^n u(-n-1) \text{ is} \quad [2009]$$

- (A) $\frac{1}{3} < |z| < \frac{1}{2}$ (B) $|z| > \frac{1}{2}$
 (C) $|z| < \frac{1}{3}$ (D) $2 < |z| < 3$

11. A system with transfer function $H(z)$ has impulse response $h(n \rightarrow n)$ defined as $h(2) = 1$, $h(3) = -1$ and $h(k) = 0$ otherwise. Consider the following statements

S_1 : $H(z)$ is a low-pass filter

S_2 : $H(z)$ is a FIR filter

Which of the following is correct? [2009]

- (A) Only S_2 is true
 (B) Both S_1 and S_2 are false
 (C) Both S_1 and S_2 are true, and S_2 is a reason for S_1
 (D) Both S_1 and S_2 are true, but S_2 is not a reason for S_1

12. Consider the z transform $X(z) = 5z^2 + 4z^{-1} + 3$; $0 < |z| < \infty$. The inverse z transform $x(n)$ is [2010]

- (A) $5\delta(n+2) + 3\delta(n) + 4\delta(n-1)$
 (B) $5\delta(n-2) + 3\delta(n) + 4\delta(n+1)$
 (C) $5u(n+2) + 3u(n) + 4u(n-1)$
 (D) $5u(n-2) + 3u(n) + 4u(n+1)$

13. The transfer function of a discrete time LTI system is given by

$$H(z) = \frac{2 - \frac{3}{4}z^{-1}}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}}$$

Consider the following statements:

S_1 : The system is stable and causal for ROC: $|z| > \frac{1}{2}$

S_2 : The system is stable but not causal for ROC: $|z| < \frac{1}{4}$

S_3 : The system is neither stable nor causal for ROC: $\frac{1}{4} < |z| < \frac{1}{2}$

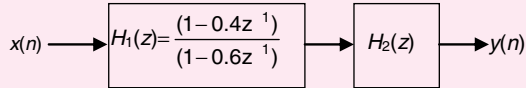
Which one of the following statements is valid? [2010]

- (A) Both S_1 and S_2 are true
 (B) Both S_2 and S_3 are true
 (C) Both S_1 and S_3 are true
 (D) S_1 , S_2 , and S_3 are all true

14. The Nyquist sampling rate for the signal $S(t) = \frac{\sin(500\pi t)}{\pi t} \times \frac{\sin(700\pi t)}{\pi t}$ is given by [2010]

(A) 400 Hz (B) 600 Hz
(C) 1,200 Hz (D) 1,400 Hz

15. Two systems $H_1(z)$ and $H_2(z)$ are connected in cascade as shown below. The overall output $y(n)$ is the same as the input $x(n)$ with a one unit delay. The transfer function of the second $H_2(z)$ is [2011]



(A) $\frac{(1-0.6z^{-1})}{z^{-1}(1-0.4z^{-1})}$ (B) $\frac{z^{-1}(1-0.6z^{-1})}{(1-0.4z^{-1})}$
(C) $\frac{z^{-1}(1-0.4z^{-1})}{(1-0.6z^{-1})}$ (D) $\frac{(1-0.4z^{-1})}{z^{-1}(1-0.6z^{-1})}$

16. If $x(n] = (1/3)^{|n|} - (1/2)^n u(n]$, then the region of convergence (ROC) of its Z transform in the Z plane will be [2012]

(A) $\frac{1}{3} < |z| < 3$ (B) $\frac{1}{3} < |z| < \frac{1}{2}$
(C) $\frac{1}{2} < |z| < 3$ (D) $\frac{1}{3} < |z|$

17. Let $y(n]$ denote the convolution of $h(n]$ and $g(n]$, where $h(n] = \left(\frac{1}{2}\right)^n u(n]$ and $g(n]$ is a causal sequence.

If $y(0] = 1$ and $y(1] = \frac{1}{2}$, then $g(1]$ equals [2012]
(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) $\frac{3}{2}$

18. A band-limited signal with a maximum frequency of 5 kHz is to be sampled. According to the sampling theorem, the sampling frequency which is not valid is [2013]

(A) 5 kHz (B) 12 kHz
(C) 15 kHz (D) 20 kHz

19. Consider two real-valued signals, $x(t]$ band-limited to $[-500 \text{ Hz}, 500 \text{ Hz}]$ and $y(t]$ band-limited to $[-1 \text{ kHz}, 1 \text{ kHz}]$. For $z(t] = x(t] \cdot y(t]$, the Nyquist sampling frequency (in kHz) is [2014]

20. Let $x(n] = \left(-\frac{1}{9}\right)^n u(n] - \left(-\frac{1}{3}\right)^n u(-n-1]$. The Region of Convergence (ROC) of the z transform of $x(n]$ [2014]

(A) is $|z| > \frac{1}{9}$ (B) is $|z| < \frac{1}{3}$
(C) is $\frac{1}{3} > |z| > \frac{1}{9}$ (D) does not exist

21. An FIR system is described by the system function

$$H(z) = 1 + \frac{7}{2}z^{-1} + \frac{3}{2}z^{-2} \quad \text{The system is} \quad [2014]$$

(A) maximum phase
(B) minimum phase
(C) mixed phase
(D) zero phase

22. Let $x(n] = x(-n]$. Let $X(z)$ be the z transform of $x(n]$. If $0.5 + j0.25$ is a zero of $X(z)$, which one of the following must also be a zero of $X(z)$. [2014]

(A) $0.5 - j0.25$ (B) $1/(0.5 + j0.25)$
(C) $1/(0.5 - j0.25)$ (D) $2 + j4$

23. The input/output relationship of a causal stable LTI system is given as $y(n] = \alpha y(n-1] + \beta x(n]$. If the impulse response $h(n]$ of this system satisfies the condition $\sum_{n=0}^{\infty} h(n] = 2$, the relationship between α and β is [2014]

(A) $\alpha = 1 - \beta/2$ (B) $\alpha = 1 + \beta/2$
(C) $\alpha = 2\beta$ (D) $\alpha = -2\beta$

24. Let $x(t] = \cos(10\pi t) + \cos(30\pi t)$ be sampled at 20 Hz and reconstructed using an ideal low-pass filter with cut-off frequency of 20 Hz. The frequency/frequencies present in the reconstructed signal is/are [2014]

(A) 5 Hz and 15 Hz only
(B) 10 Hz and 15 Hz only
(C) 5 Hz, 10 Hz, and 15 Hz only
(D) 5 Hz only

25. For an all pass system

$$H(z) = \frac{(z^{-1} - b)}{(1 - az^{-1})},$$

where $|H(e^{j\omega})| = 1$, for all ω . If $\text{Re}(a) \neq 0$, $\text{Im}(a) \neq 0$, then b equals [2014]

(A) a (B) a^*
(C) $1/a^*$ (D) $1/a$

26. Let $H_1(z) = (1 - pz^{-1})^{-1}$, $H_2(z) = (1 - qz^{-1})^{-1}$, $H(z) = H_1(z) + rH_2(z)$. The quantities p, q, r are real numbers.

Consider $p = \frac{1}{2}$, $q = -\frac{1}{4}$, $|r| < 1$. If the zero of $H(z)$ lies on the unit circle, then $r =$ [2014]

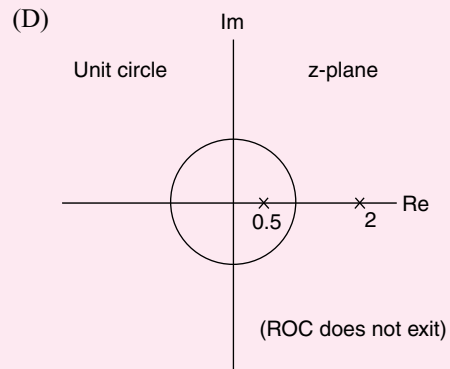
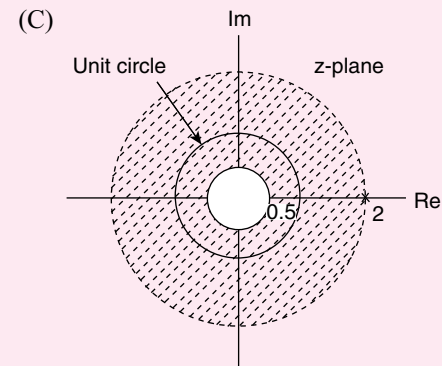
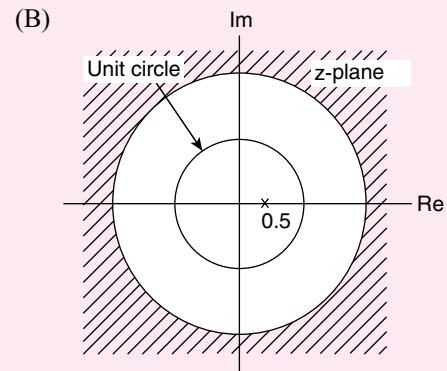
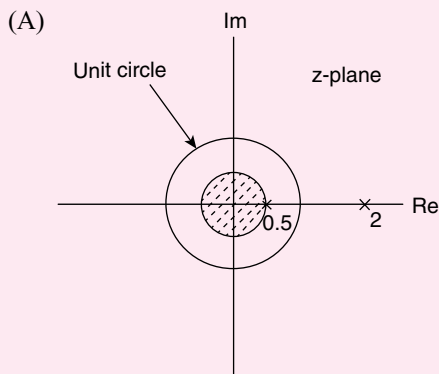
27. The z transform of the sequence $x(n]$ is given by $X(z) = \frac{1}{(1 - 2z^{-1})^2}$, with the region of convergence $|z| > 2$.

Then, $x(2)$ is [2014]

28. The sequence $x(n] = 0.5^n u(n]$, where $u(n]$ is the unit step sequence, is convolved with itself to obtain $y(n]$.

Then, $\sum_{n=-\infty}^{+\infty} y(n]$ is [2014]

29. Two causal discrete-time signals $x[n]$ and $y[n]$ are related as $y[n] = \sum_{m=0}^n x[m]$. If the z -transform of $y[n]$ is $\frac{2}{z(z-1)^2}$, the value of $x[2]$ is _____. [2015]
30. Consider the sequence $x[n] = a^n u[n] + b^n u[1]$, where $u[n]$ denote the unit step sequence and $0 < |a| < |b| < 1$. The region of convergence (ROC) of the z transform of $x[n]$ is [2016]
- (A) $|z| > |a|$ (B) $|z| > |b|$
 (C) $|z| < |a|$ (D) $|a| < |z| < |b|$
31. A discrete-time signal $x[n] = \delta[n-3] + 2\delta[n-5]$ has z -transform $X(z)$. If $Y(z) = X(-z)$ is the z -transform of another signal $y[n]$, then [2016]
- (A) $y[n] = x[n]$ (B) $y[n] = x[-n]$
 (C) $y[n] = -x[n]$ (D) $y[n] = -x[-n]$
32. The ROC (region of convergence) of the z -transform of a discrete time signal is represented by the shaded region in the z -plane. If the signal $x[n] = (2.0)^{|n|}$, $-\infty < n < +\infty$, then the ROC of its z -transform is represented by [2016]



ANSWER KEYS

EXERCISES

Practice Problems 1

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. B | 2. A | 3. C | 4. B | 5. C | 6. D | 7. A | 8. B | 9. B | 10. C |
| 11. D | 12. C | 13. A | 14. B | 15. A | 16. A | 17. A | 18. D | 19. C | 20. D |
| 21. B | 22. B | | | | | | | | |

Practice Problems 2

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. A | 3. B | 4. D | 5. B | 6. D | 7. B | 8. B | 9. C | 10. B |
| 11. D | 12. B | 13. D | 14. A | 15. A | 16. D | 17. B | 18. C | 19. B | 20. B |
| 21. A | | | | | | | | | |

Previous Years' Questions

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|----------|--------|------------------|-------|
| 1. D | 2. C | 3. C | 4. C | 5. D | 6. C | 7. B | 8. B | 9. C | 10. A |
| 11. A | 12. A | 13. C | 14. C | 15. B | 16. C | 17. A | 18. A | 19. 2.99 to 3.01 | |
| 20. C | 21. C | 22. B | 23. A | 24. A | 25. B | 26. -0.5 | 27. 12 | 28. 4 | 29. 0 |
| 30. B | 31. C | 32. D | | | | | | | |